Predicting Product Launch Success on Product Hunt Using Multimodal Text and Metadata Features

Author: Omar Abdellall

Course: Intro to Machine Learning

1) Problem Statement

Predict launch success using only pre-launch signals.

Tasks: top-20 percent classification and upvote regression.

2) Data Collection and Coverage

Total samples: 1800

Date range: 2022-01-29 08:00:00+00:00 to 2025-12-31 08:01:00+00:00

Unique launch days: 51

Positive rate (is_top20): 20.33%

Year counts: 2022=474, 2023=480, 2024=479, 2025=367

3) Methodology

Features: metadata + TF-IDF + Sentence-BERT embeddings

Models: Logistic/XGBoost (classification), Linear/Ridge/XGBoost (regression)

Split: 80/20 stratified, 5-fold CV for model selection

Regression robustness: log1p target training with inverse-transform

4) Results

Classification metrics:

logreg_metadata_tfidf: AUC=0.727, F1=0.404, P=0.330, R=0.521
logreg_metadata_sbert: AUC=0.720, F1=0.404, P=0.315, R=0.562
logreg_metadata_only: AUC=0.717, F1=0.394, P=0.312, R=0.534
xgb_classifier_metadata_sbert: AUC=0.645, F1=0.257, P=0.464, R=0.178

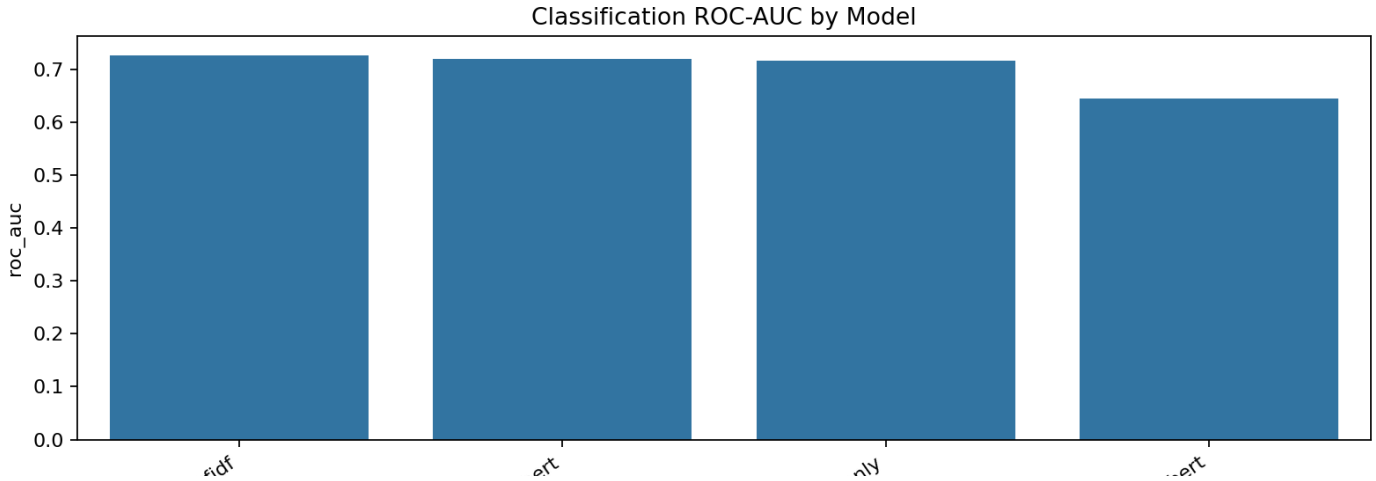
Regression metrics:

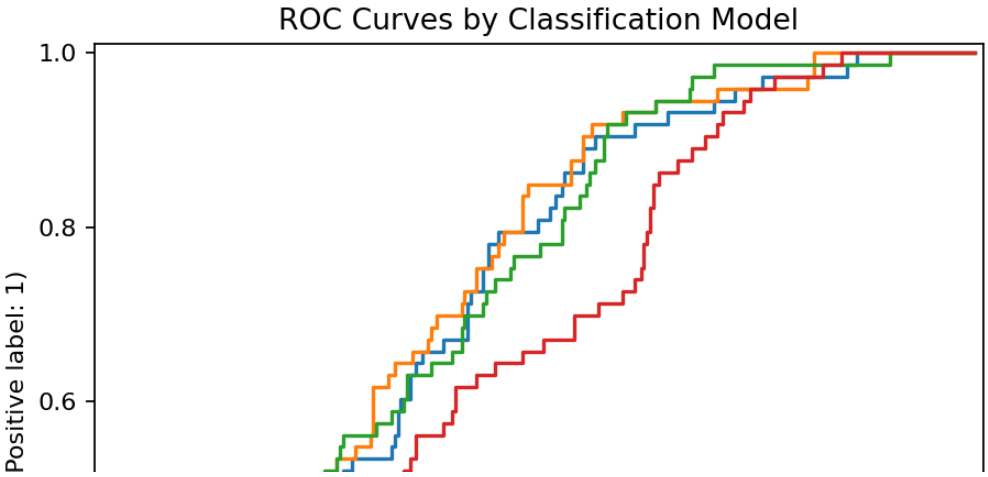
xgb_regressor_metadata_sbert: R2=0.111, MAE=75.07, R2_log=0.347
daily_mean_baseline: R2=0.054, MAE=90.36, R2_log=-0.066
ridge_metadata_tfidf: R2=0.037, MAE=78.56, R2_log=0.280
ridge_metadata_sbert: R2=0.035, MAE=79.13, R2_log=0.269
linear_metadata_only: R2=-0.038, MAE=80.04, R2_log=0.261

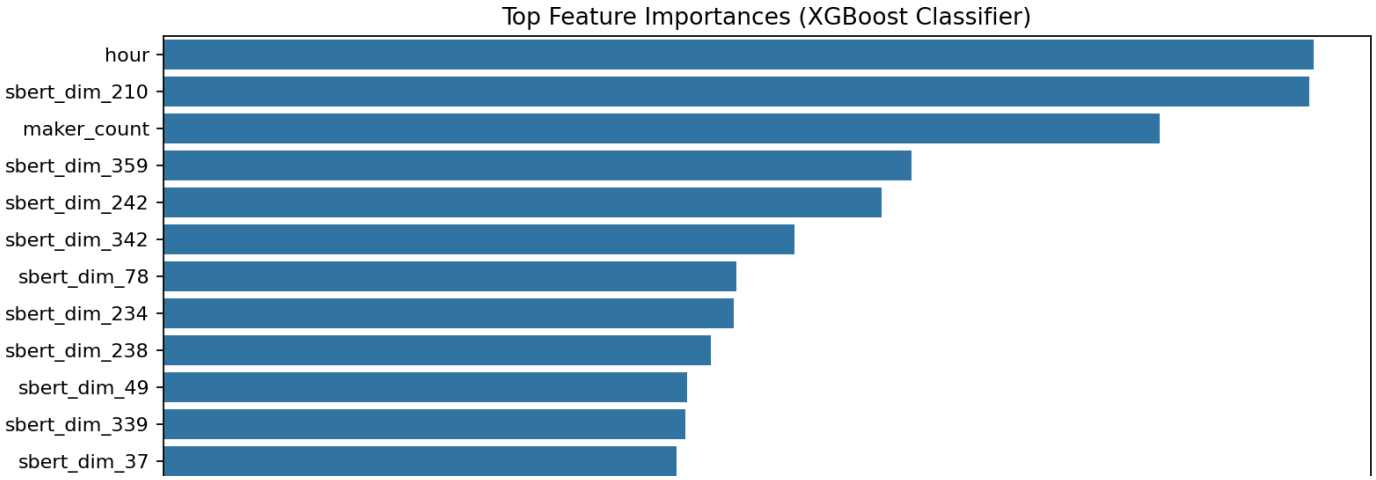
Target check:

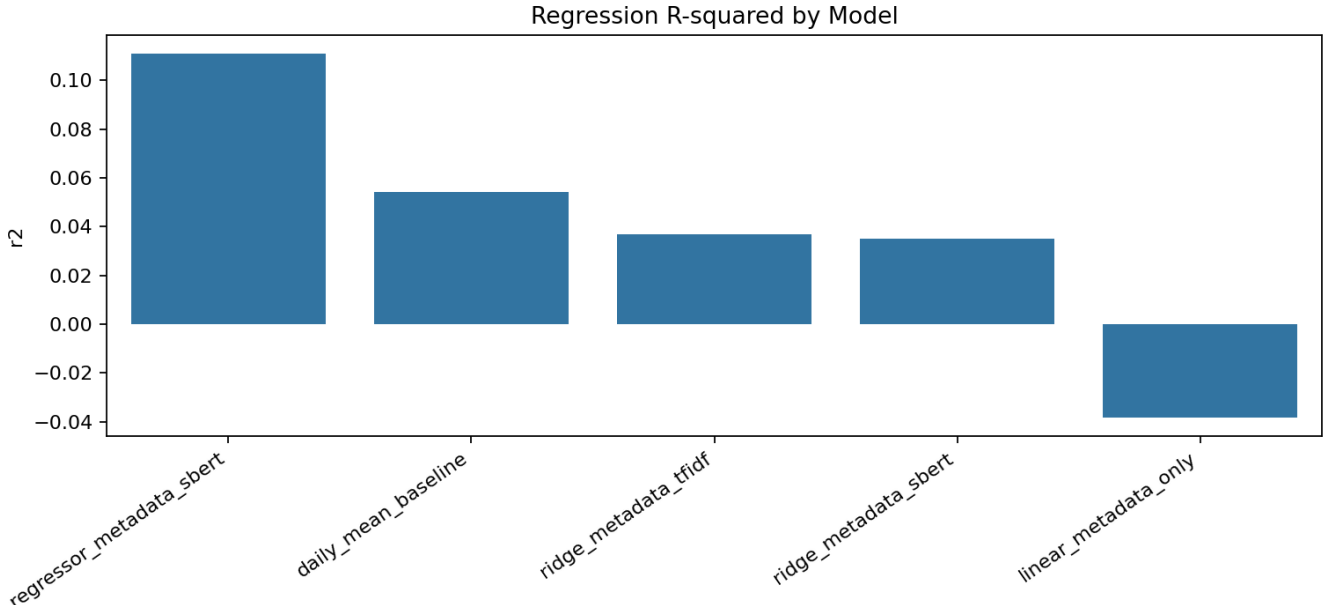
Classification target met (AUC >= 0.70).
Original-scale regression target not yet met (R2 < 0.30).

5) Key Figures









6) Conclusion and Next Steps

The pipeline is fully runnable and delivers strong classification performance.
Regression on original vote scale remains challenging due heavy-tailed targets.
Most impactful next step: continue deeper collection with monthly windows and rerun the same training/evaluation stack to improve original-scale R2.